

SET 2 – Complete Answers

Q1. Algorithms – Merge Sort (5 Marks)

Using AI assistance, the following Python code implements Merge Sort and runs it on the given list:

```
def merge(left, right):
    i = j = 0
    merged = []
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            merged.append(left[i])
            i += 1
        else:
            merged.append(right[j])
            j += 1
    if i < len(left):
        merged.extend(left[i:])
    if j < len(right):
        merged.extend(right[j:])
    return merged

def merge_sort(arr):
    if len(arr) <= 1:
        return arr[:]
    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])
    return merge(left, right)

data = [45, 12, 3, 67, 34, 21]
sorted_data = merge_sort(data)
print(sorted_data)
```

Output of the program: [3, 12, 21, 34, 45, 67]

Time Complexity Explanation

Merge Sort works by repeatedly dividing the list into halves and merging them. The time complexity is $O(n \log n)$ in the best, average, and worst cases.

How AI Improved Correctness

AI ensured correct indentation, recursion structure, merging logic, and validated the implementation. It helped avoid human mistakes such as incorrect merge conditions or missing base cases.

Q2. Web Frontend – Student Registration Form (5 Marks)

A webpage with Name, Email, Course fields, Submit button, and JavaScript validation:

```
<!doctype html>
<html>
<head>
<title>Student Registration</title>
</head>
<body>
<form id="regForm">
Name: <input type="text" id="name"><br>
Email: <input type="email" id="email"><br>
Course:
<select id="course">
<option value="">Select a course</option>
<option>Computer Science</option>
<option>Mathematics</option>
<option>Physics</option>
</select><br>
<button type="submit">Submit</button>
</form>

<script>
document.getElementById("regForm").addEventListener("submit", function(e){
  e.preventDefault();
  let name = document.getElementById("name").value.trim();
  let email = document.getElementById("email").value.trim();
  let course = document.getElementById("course").value;

  if(!name){ alert("Enter name"); return; }
  if(!email){ alert("Enter email"); return; }
  if(!course){ alert("Select a course"); return; }

  alert("Registration Successful!");
});
</script>
</body>
```

</html>